Cambridge Assessment International Education

# Cambridge IGCSE®

**BIOLOGY** **0610/04**

Paper 4 Theory (Extended) **For examination from 2020**

MARK SCHEME

Maximum Mark: 80

**Specimen**

This document consists of **8** printed pages.

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mark scheme abbreviations

| | |
|---|---|
| ; | separates marking points |
| / | alternative responses for the same marking point |
| not | do not allow |
| allow | accept the response |
| ecf | error carried forward |
| avp | any valid point |
| ora | or reverse argument |
| owtte | or words to that effect |
| underline | actual word given must be used by candidate (grammatical variants excepted) |
| ( ) | the word / phrase in brackets is not required but sets the context |
| max | indicates the maximum number of marks |
| Any [number] from: | accept the [number] of valid responses |
| note: | additional marking guidance |

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**1** **(a)** animals written in the correct boxes in the food web

vultures;

cheetahs;

mice / mouse; [3]

**(b)** (primary) producer;

primary consumer; [2]

**(c)** **(i)** Sun; [1]

**(ii)** (lost) to the atmosphere / (lost as) infra-red (radiation) / heat / owtte; [1]

**(d)** Any three from:

idea that small percentage of energy from Sun is 'fixed' by photosynthesis /

most energy from Sun not available / reference to wrong wavelength;

energy is lost between and within trophic levels;

ref. to 10% energy transfer / ora (per trophic level);

note: if magnitude given, e.g. '90% lost between trophic levels', award 2 marks

ref. to material that is inedible or indigestible;

ref. to (small) total percentage reaching fourth trophic level (cumulative idea);

not enough energy in fourth trophic level to support another level;

avp; [max 3]

**(e)** Any three from:

feed is expensive;

more energy efficient to feed humans on crops or producers or animals that are used to make the (fish) food;

waste feed causes eutrophication of water supplies;

diseases or parasites spread easily (in captivity);

diseases spread to other organisms in the wild;

chemicals used to control disease are also pollutants (e.g. antibiotics);

avp; e.g. animal welfare concerns [max 3]

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**2** **(a)** Any three from:

muscular contraction / movement / pump blood;

allow: maintain posture

maintenance of body temperature;

active transport described / example such as nerve impulses;

metabolic reactions / named example (e.g. excretion / biosynthesis / digestion);

mitosis / nuclear division / cell division;

growth / replacement / repair;

making gametes / owtte;

avp; [max 3]

**(b)** **(i)** respiration; [1]

**(ii)** $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ [1]

**(c)** Any four from:

takes time for;

oxygen debt (to be repaid);

more oxygen needed;

lactic acid / lactate;

builds up in muscles / needs to be cleared away;

lactic acid is broken down / respired; [max 4]

**(d)** Any five from:

as body temperature increases;

<u>vasodilation</u>;

(relaxation / owtte) of arterioles;

allow: arteries

increase in supply of blood to skin capillaries;

(causes) loss of heat;

by, conduction / convection / radiation;

increase in blood flow to sweat glands;

increased production of sweat;

loss of heat by evaporation; [max 5]


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**3** **(a)** ability to detect and respond to changes in the environment; [1]

**(b)** correct label and name of:
sensory neurone;
relay / connector / intermediate neurone;
motor / effector neurone; [3]

**(c)** automatic / no thought required / not a conscious action;
allow: no (higher centres in) brain involved
ignore: reference to speed of response
stimulus always leads to the same response;
allow: fixed response [2]

**(d)** **(i)** a junction between two neurones; [1]

**(ii)** Any three from:
(an impulse triggers) the release of neurotransmitters (into the gap);
diffuse across the gap;
binds to receptors;
which allows the passage of the impulse; [max 3]

**(e)** Any two from:
heart beats faster / increased pulse;
increased rate of breathing;
stimulates breakdown of glycogen in the liver;
increases blood glucose concentration;
dilate pupils;
heightened sensitivity / increased mental awareness / owtte;
allow: sharper senses / more alert / owtte [max 2]

**(f)** Any one from:
nervous control is faster / ora;
hormonal control is longer-lasting / ora;
note: comparison must be made [max 1]

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**4** **(a)** **R and Y**;
**RY**;
orange; [3]

**(b)**

| | genotypes of offspring |
|---|---|
| cross 2 | **RR**, **YY**, **RY** |
| cross 3 | **RR**, **RY** |
| cross 4 | **YY**, **RY** |

allow: ecf from **4(a)** [3]

**(c)** Any three from:
phenotype of **RY** (offspring of **cross 1**) is different from either parent or the homozygous genotypes / owtte;
the phenotype was intermediate / mixture of two colours;
offspring of **cross 2** gives three phenotypes not two;
offspring of **crosses 3** and **4** both give two phenotypes;
if dominance **cross 3** or **4** would give one phenotype only;
allow: incomplete dominance
allow: both alleles are expressed [max 3]

**(d)** Any two from:
transfer of pollen from anthers or stamen to stigma;
self = within same flower (or flower on same plant);
cross = between flowers on different plants (of same species); [max 2]

**(e)** Any four from:
limited variation;
offspring become homozygous (over time) / owtte;
allow: reference to inbreeding / limited gene pool
variation is due to mutation;
low chance that mutations will be expressed / owtte;
offspring will be well adapted to conditions near parent;
if environment does not change;
limited opportunity for evolution if environment changes / will not be able to adapt to change in the environment;
allow: reference to disease in context (as a change)
avp; e.g. some variation due to reassortment of chromosomes and crossing over during meiosis / reduced variation leads to intraspecific competition locally; [max 4]


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**5** **(a)** *water jacket*

Any four from:

maintain optimum / constant temperature;

allow: prevent overheating

to prevent enzymes denaturing;

(because as) fungus respires;

releases heat so temperature in the fermenter increases;

which would kill fungus;

(therefore) no product / no penicillin / owtte;

*addition of acids and alkalis*

Any two from:

maintains pH / keeps pH constant;

enzymes need optimum pH;

to give maximum enzyme activity / rate of reaction at its fastest;

to give maximum yield / owtte;

allow: stop enzymes denaturing

[max 6]

**(b)** **(i)** 40–50; [1]

**(ii)** mitosis; [1]

**(iii)** Any three from:

nutrients are used up;

limiting (factors);

explanation of limiting factor;

allow: factor in shortest supply / owtte

waste products accumulate;

wastes are toxic;

penicillin could inhibit growth;

population reaches carrying capacity;

avp; [max 3]

**(c)** **(i)** fungus grows when no penicillin produced;

during first 20 hours; [2]

**(ii)** Any one from:

no more growth of fungus / fungus is dead;

no further production of penicillin / no advantage in continuing; [max 1]


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**(d)** Any three from:
purifying or separating penicillin;
from waste or toxins / owtte;
concentration;
making into pills / owtte;
avp; e.g. colour / taste [max 3]

**(e)** Any two from:
viruses have no metabolism;
allow: viruses do not have ribosomes
idea that viruses have no target for antibiotics / owtte;
antibiotics stop cell wall growth;
viruses have no cell wall;
antibiotics stop enzymes working; [max 2]

**6** **(a)** **A** epithelium / epithelial lining;
**B** lacteal;
**C** capillary / blood vessel; [3]

**(b)** Any three from:
*microvilli*
increases / large surface area;
for absorption;
allow: diffusion / active transport (into villus)
*mitochondria*
(for) respiration;
provide energy / ATP;
for active uptake / transport; [max 3]

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